

Jie (Sophia) Gao

Malone Postdoctoral Fellow
Malone Center for Engineering in Healthcare
Center for Language and Speech Processing
Johns Hopkins University

6225 Smith Ave, The Johns Hopkins at Mt. Washington campus, Baltimore, MD 21209, USA

jgao77@jh.edu
[gaojie058.github.io](https://github.com/gaojie058)
+1 (410) 710-0555

Research Interests

Human-Centered Agentic AI · Human-AI Collaboration · AI-assisted Qualitative Analysis · Trustworthy AI · Agent Behavior Analysis · Agent Oversight and Evaluation

Academic Employment

Malone Postdoctoral Fellow	Mar 2025 – Present
Johns Hopkins University, Baltimore, MD, USA	
Advisors: Mark Dredze, Ziang Xiao, Chien-Ming Huang	
Postdoctoral Associate	Feb 2024 – Feb 2025
Singapore–MIT Alliance for Research and Technology (SMART), MIT, Singapore	
Advisor: Thomas W. Malone	

Education

Ph.D., Information Systems Technology and Design	Sep 2019 – Jan 2024
Singapore University of Technology and Design, Singapore	
Advisor: Simon Perrault	
Thesis: “Tool Support for Human-AI Collaboration in Qualitative Data Analysis.”	
— Visiting PhD Student, University of Notre Dame, USA (Host: Toby Jia-Jun Li)	Spring 2023
— Visiting PhD Student, National University of Singapore (Host: Shengdong Zhao)	Summer and Fall 2022
M.Eng., Software Engineering / Information Product Design	Sep 2017 – Jun 2019
Zhejiang University, Hangzhou, China · Admitted without entrance examination	
B.Eng., Digital Media Technology	Sep 2013 – Jun 2017
Shanghai University, Shanghai, China · Ranked 1st in major	

Selected Awards & Honors

ACM SIGSOFT Distinguished Paper Award, ICPC 2026	2026
Malone Postdoctoral Fellowship, Johns Hopkins University	2024
SUTD PhD Fellowship, Singapore University of Technology and Design	2019
Excellent Graduate Student, Zhejiang University	2019
Excellent Undergraduate Student, Shanghai University	2017
Innovation and Entrepreneurship Scholarship, Shanghai University	2016
Leadership Scholarship, Shanghai University	2016

Selected Publications

* denotes co-first author, † denotes corresponding and senior author.

Peer-Reviewed Publications

1. **Jie Gao**, Kaiser Sun, Jen-tse Huang, Katherine Van Koeveering, Sijie Ji, Heyuan Huang, Weiyan Shi, Zhuoran Lu, Ziang Xiao, Daniel Khashabi, Mark Dredze. “How to Interpret Agent Behavior.” *Conference on Neural Information Processing Systems (NeurIPS 2026)*.
2. Zhimin Lin, Kun Cheng, Zhiyao Shu, Junhua Fang, Juntao Li, Fan Bai, **Jie Gao**†. “Agent-as-Peer-Debrief: A Multi-Agent Framework with Perspective-Based Refinement for Qualitative Analysis.” *Proceedings of the 2026 Conference on Empirical Methods in Natural Language Processing (EMNLP 2026, Main Conference/Oral)*.
3. **Jie Gao**, Yue Xue, Xiaofei Xie, Junming Cao, SoeMin Thant, Erika Lee, Bowen Xu. “Understanding Codebase like a Professional! Human–AI Collaboration for Code Comprehension.” *The 34th IEEE/ACM International Conference on Program Comprehension (ICPC 2026)*. (ACM SIGSOFT Distinguished Paper Award)
4. **Jie Gao**, Yuchen Guo, Gionnieve Lim, Tianqin Zhang, Zheng Zhang, Toby Jia-Jun Li, Simon Perrault. “CollabCoder: A Lower-barrier, Rigorous Workflow for Inductive Collaborative Qualitative Analysis with Large Language Models.” *Proceedings of the ACM Conference on Human Factors in Computing Systems (CHI 2024)*.
5. **Jie Gao**, Kenny Tsu Wei Choo, Junming Cao, Roy Ka Wei Lee, Simon Perrault. “CoAICoder: Examining the Effectiveness of AI-assisted Human-to-Human Collaboration in Qualitative Analysis.” *ACM Transactions on Computer-Human Interaction (TOCHI 2023)*.
6. **Jie Gao***, Simret Araya Gebreegziabher*, Kenny Tsu Wei Choo, Toby Jia-Jun Li, Simon Tangi Perrault, Thomas W Malone. “A Taxonomy for Human-LLM Interaction Modes: An Initial Exploration.” *Proceedings of the ACM Conference on Human Factors in Computing Systems (CHI Late-Breaking Work 2024)*.
7. **Jie Gao**, Xiayin Ying, Junming Cao, Yifan Yang, Pin Sym Foong, Simon Perrault. “Differences of Challenges of Working from Home (WFH) between China and the US during COVID-19.” *Proceedings of the ACM Conference on Human Factors in Computing Systems (CHI Late-Breaking Work 2022)*.
8. Kaiser Sun, Bernal Jiménez Gutiérrez, Hongjun Liu, Jingyu Zhang, **Jie Gao**, Mark Dredze, Daniel Khashabi. “Accurate but Not Humble: Evaluating Epistemic Humility in LLM Agents under Knowledge Conflict.” *Findings of the Association for Computational Linguistics: EMNLP 2026*.
9. Zhuoran Lu, Yangyang Yu, Zhuoyan Li, Yibo Meng, Nan Jiang, Chengxi Zang, **Jie Gao**, Ziang Xiao. “Using Grounded Theory for Agent Behavior Analysis at Scale.” *Findings of the Association for Computational Linguistics: EMNLP 2026*.
10. Marianne Aubin Le Quéré, Hope Schroeder, Casey Randazzo, **Jie Gao**. “The State of Large Language Models in HCI Research: Workshop Report.” *ACM Interactions, 2025*.
11. ShunYi Yeo, Gionnieve Lim, **Jie Gao**, Weiyu Zhang, Simon Tangi Perrault. “Help Me Reflect: Leveraging Self-Reflection Interface Nudges to Enhance Deliberativeness on Online Deliberation Platforms.” *Proceedings of the ACM Conference on Human Factors in Computing Systems (CHI 2024)*.

12. Zheng Zhang, **Jie Gao**, Ranjodh Singh Dhaliwal, Toby Jia-Jun Li. “VISAR: A Human-AI Argumentative Writing Assistant with Visual Programming and Rapid Draft Prototyping.” *Proceedings of the ACM Symposium on User Interface Software and Technology (UIST 2023)*.
13. Nuwan Janaka, **Jie Gao**, Lin Zhu, Shengdong Zhao, Lan Lyu, Peisen Xu, Maximilian Nabokow, Silang Wang, Yanch Ong. “GlassMessaging: Towards Ubiquitous Messaging Using OHMDs.” *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies (IMWUT 2023)*.
14. Junming Cao, Bihuan Chen, Longjie Hu, **Jie Gao**, Kaifeng Huang, Xuezhi Song, Xin Peng. “Understanding Complexity and Its Impact on Testing in ML-Enabled Systems.” *IEEE International Conference on Software Maintenance and Evolution (ICSME 2023)*.

Workshops, Demos & Posters

1. **Jie Gao**, Yuchen Guo, Toby Jia-jun Li, Simon Perrault. “CollabCoder: A GPT-Powered Workflow for Collaborative Qualitative Analysis.” *CSCW Demo 2023*.
2. **Jie Gao**, Leijing Zhou, Miaomiao Dong, Fan Zhang. “Expressive Plant: A Multisensory Interactive System for Sensory Training of Children with Autism.” *UbiComp Poster 2018*.
3. Ruyuan Wan, Haonan Wang, Ting-Hao ‘Kenneth’ Huang, **Jie Gao**†. “From Noise to Nuance: Enriching Subjective Data Annotation through Qualitative Analysis.” *The 4th HCI+NLP Workshop at EMNLP 2025*.

Invited Talks & Presentations

“Toward Seeing What Agents Do: Interpretive Work in the Age of Agentic AI” Monash University, Australia	Jun 2026
“Toward Seeing What Agents Do: Interpretive Work in the Age of Agentic AI” Fudan University, China	May 2026
“Human-AI Collaboration for Complex Unstructured Data Analysis” Western Michigan University, USA	Nov 2025
“From Automation to Human-AI Collaboration: Methods and Applications for Insights Extraction in Unstructured Data” Malone Center Research Lunch, Johns Hopkins University, USA	Oct 2025
“AI for Text Production and Analysis” HCIX Research Club	Jul 2025
“Designing Human-AI Teams to Extract Insights from Unstructured Data” HCI Seminar, Johns Hopkins University, USA	Apr 2025
“Human-Centric Software Engineering in the Era of Large Language Models” Singapore Management University, Singapore	Mar 2025
“Extracting Insights from (Un)structured Data: Methods and Applications” M3S Seminar, Singapore-MIT Alliance for Research and Technology, Singapore	Feb 2025

Professional Leadership

Workshop Organization

Lead Organizer, “Interpreting Agent Behavior (IAB): The First Workshop on Human-Centered Interpretation for Understanding Agents, Humans, and Interaction,” NeurIPS 2026 Workshop 2026

Co-organizer, “LLMs as Research Tools: Applications and Evaluations in HCI Data Work,” CHI 2024 Workshop 2024

Associate Chair / Area Chair

ACM Conference on Human Factors in Computing Systems (CHI), Main Track

Associate Chair (incoming) 2027

Associate Chair 2026

Associate Chair, Best Paper Committee 2025

ACM Conference on Human Factors in Computing Systems (CHI), Late-Breaking Work Track

Associate Chair 2023

Associate Chair 2022

Teaching Experience

Guest Lecture, *Advanced HCI*, Johns Hopkins University Apr 2026

Guest Lecture, *NLP for Computational Social Science*, Johns Hopkins University Apr 2026

Instructor, *Claude Code: How-To Session*, Data Science and AI Institute (DSAI), Johns Hopkins University Spring 2026

Attracted 40+ attendees across faculty, graduate, and undergraduate levels.

Guest Lecture, Public health course May 2025

Teaching Assistant, *50.002: Introduction to Algorithms*, Singapore University of Technology and Design Fall 2020

Teaching Assistant, *50.017: Graphics and Visualization*, Singapore University of Technology and Design Summer 2020

Mentoring Experience

Zhimin Lin (Master, NLP Group, Soochow University) Spring 2026

Yifan Zhang (PhD Student, National University of Singapore) Spring 2026

Haonan Wang (Master, Johns Hopkins University) Summer–Fall 2025

Ruyuan Wan (PhD Student, Pennsylvania State University) Summer 2024

Zhiyao Shu (Master Research Intern, UC Berkeley) Summer 2024 – Spring 2025

Xiaoxian Zhang (Master Research Intern, National University of Singapore) Summer 2024

Erika Lee (Undergraduate Research Intern, UC San Diego) Summer 2024

Yuchen Guo (Undergraduate Research Intern, SUTD) Fall 2022

Tianqin Zhang (Undergraduate Research Intern, SUTD) Fall 2022

Professional Service

Invited Reviewer

Conference on Neural Information Processing Systems (NeurIPS)	2026
Nordic Conference on Human–Computer Interaction (NordiCHI)	2026
International Journal of Human–Computer Interaction (IJHCI)	2025
Empirical Software Engineering (EMSE)	2025
ACM Symposium on User Interface Software and Technology (UIST)	2025
Computers in Human Behavior Reports	2025
ACM Transactions on Interactive Intelligent Systems (TiiS)	2024–2025
ACM Conference on Human Factors in Computing Systems (CHI)	2023–2024
International Conference on Artificial Intelligence in Education (AIED)	2023
ACM Conference on Computer-Supported Cooperative Work (CSCW)	2022–2023

Updated as of September 2026